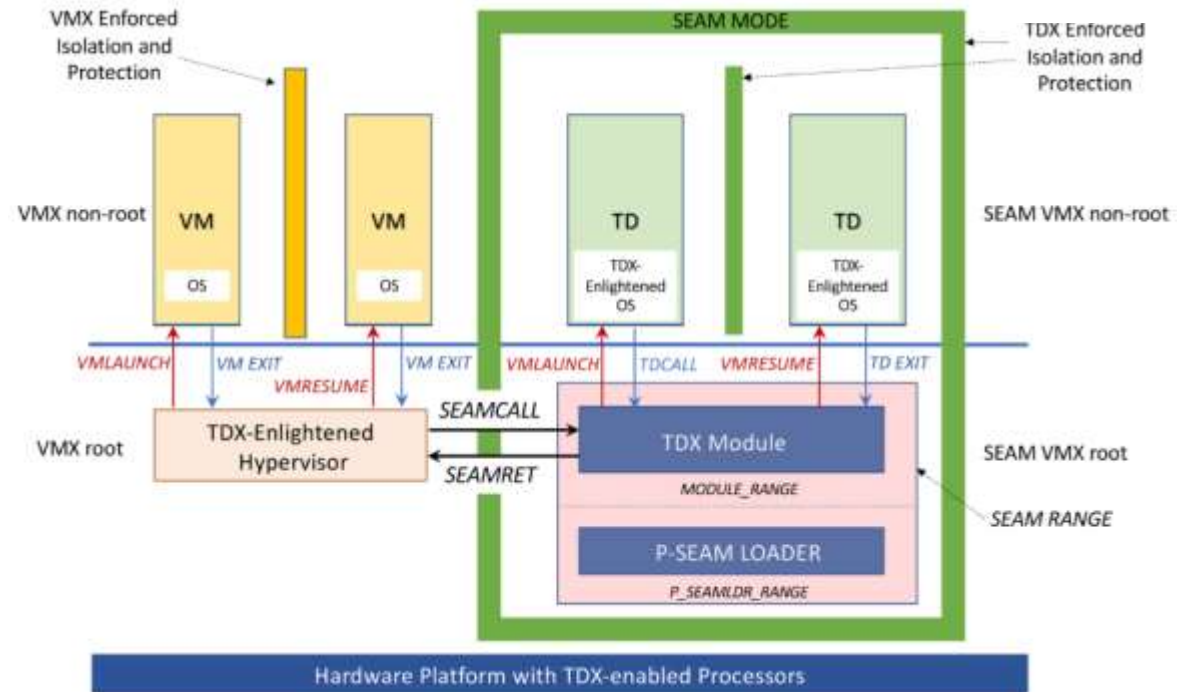
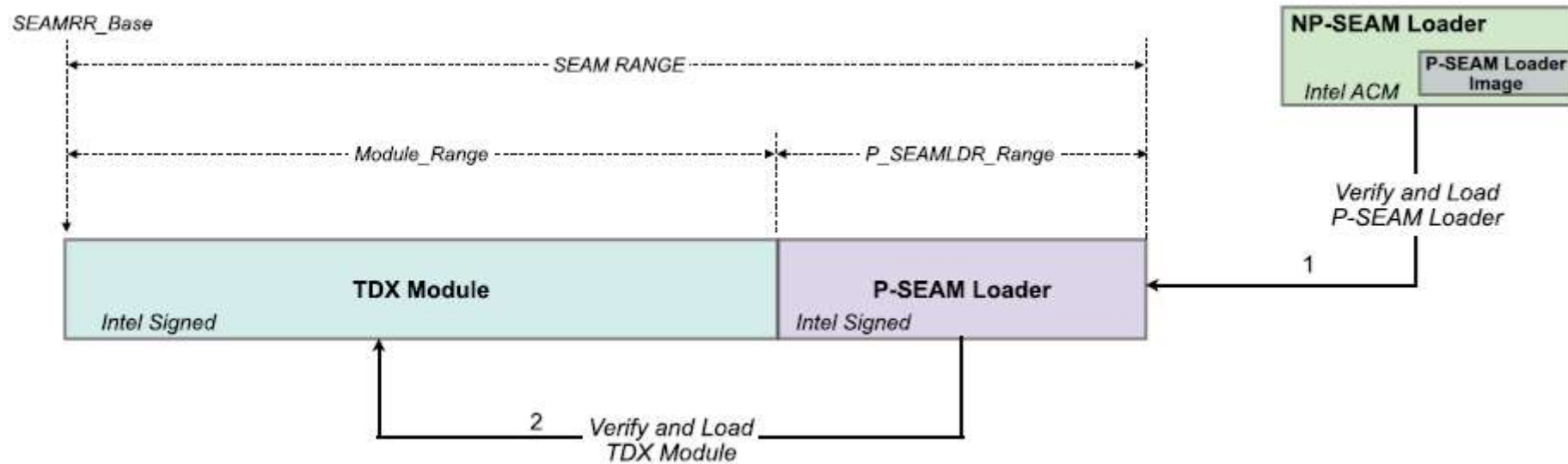


TDX Module

- TDX Module is an Intel-signed and CPU-attested software module that leverages the features of TDX-enabled processors to facilitate the construction, execution, and termination of TDs while enforcing the security guarantees.
- It provides two sets of interface functions:
 - Host-side interface functions for a TDX-enlightened hypervisor
 - Guest-side interface functions for TDs.
- It is loaded and executed in the SEAM Range, which is a portion of system memory

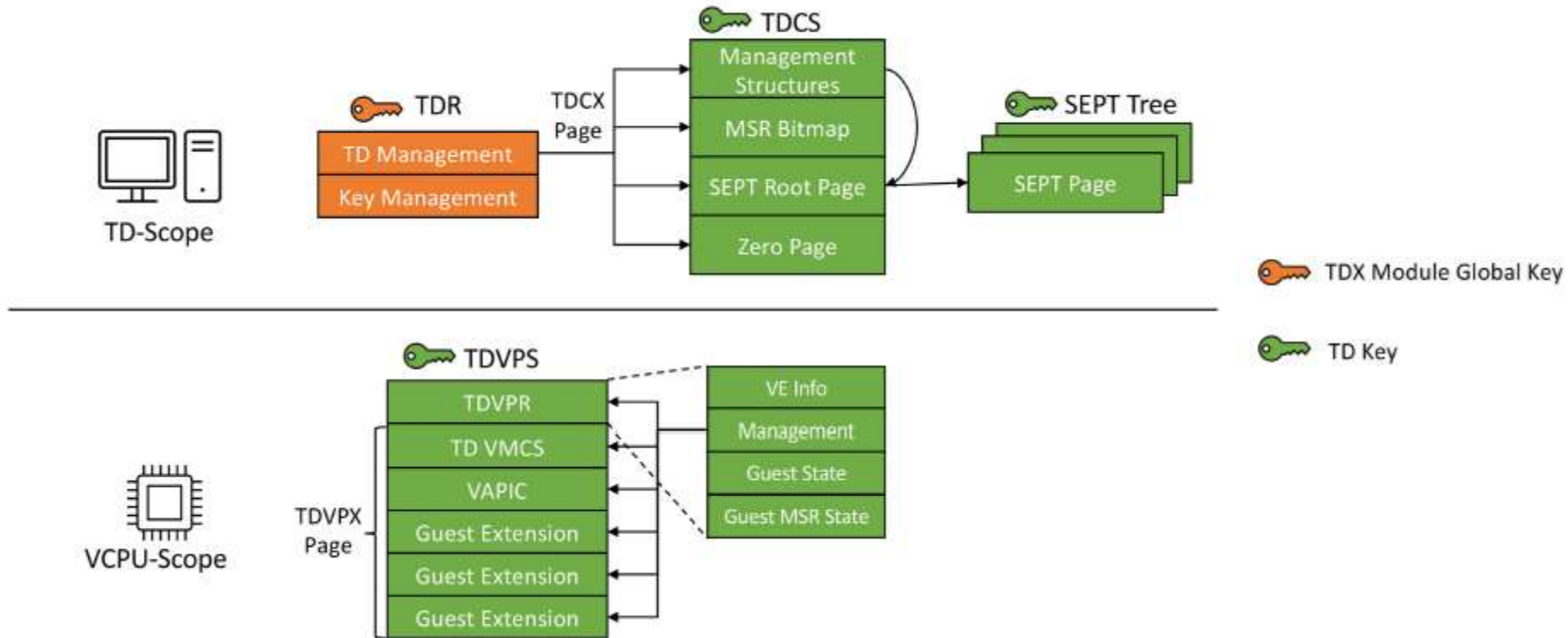


TDX Module Loading



- Intel Non-Persistent SEAM Loader (NP-SEAM Loader)
 - Intel Authenticated Code Module (ACM).
 - ACMs are Intel-signed modules that run within the internal RAM of the processor.
 - The NP-SEAM Loader is authenticated and loaded by the Intel Trusted Execution Technology (TXT)

TD Metadata



- The TDX Module is responsible for managing the entire life cycle of TDs and maintaining metadata for each TD instance.
- The TDX Module ensures that the memory encryption is applied to the metadata to prevent the hypervisor from accessing or modifying it.

Metadata Structures

TDR (Trust Domain Root)

Initial structure with encryption key info and TDCX references, protected by global private key

TDCS (Control Structure)

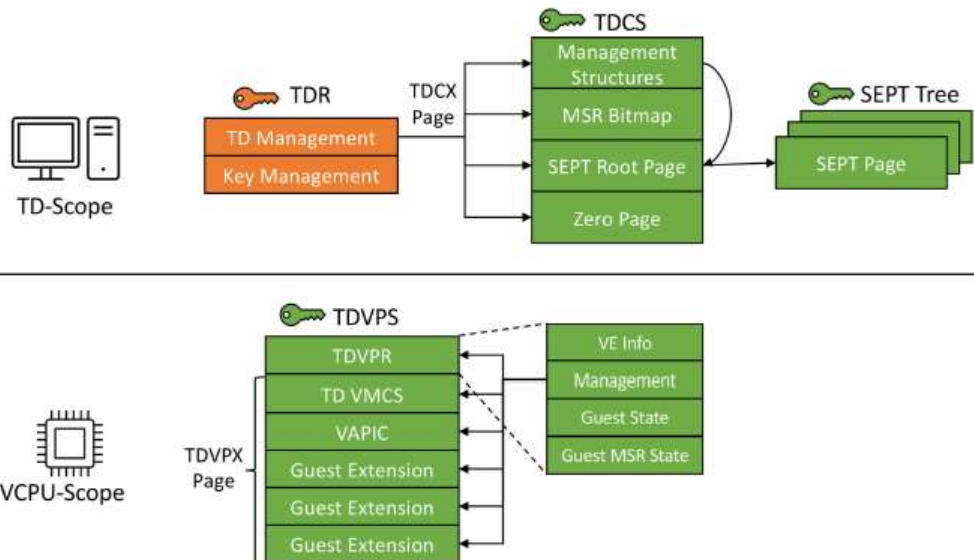
Four pages for TD management, MSR bitmaps, SEPT root, encrypted with TD private key

TDVPS (Virtual Processor State)

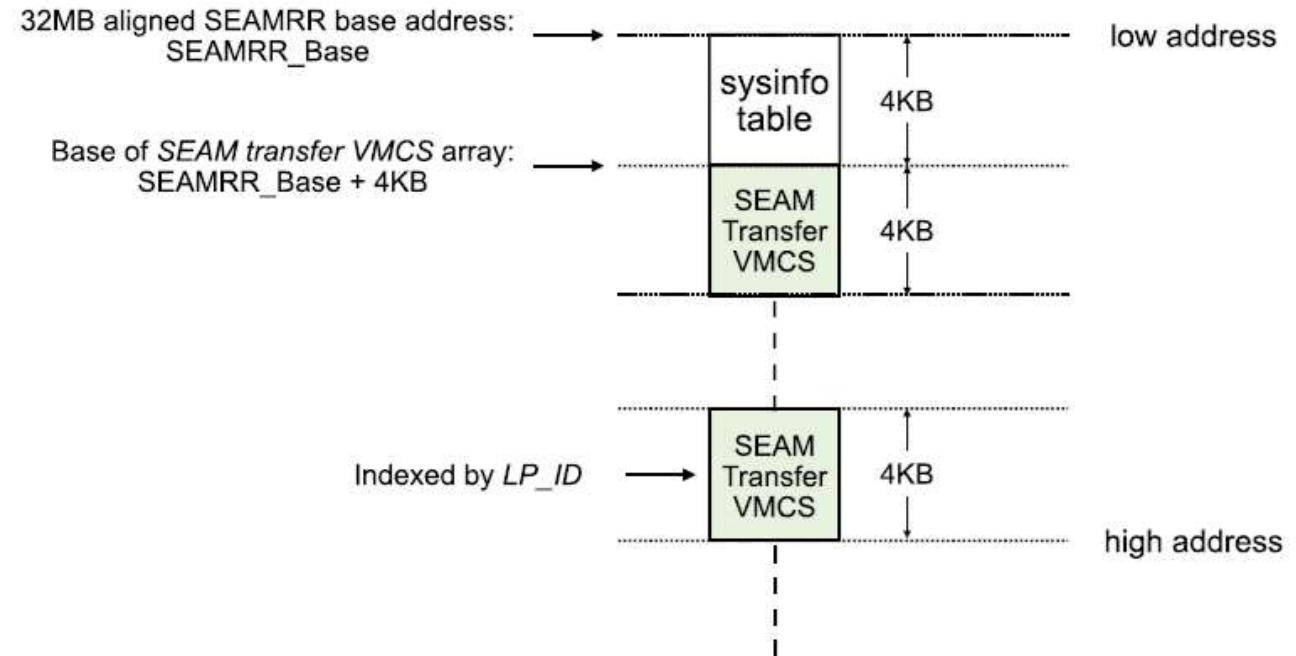
Six pages per virtual CPU: TDVPR, TD Transfer VMCS, VAPIC, guest extensions

SEPT (Secure EPT)

Protected page table for TD private memory address translation



Context Switches



■ Hypervisor ↔ TDX Module

- In TDX, a hypervisor is required to interact with the TDX Module for managing TDs through SEAMCALL interface functions.
- When SEAMCALL is made, the processor transitions from the VMX root mode to the SEAM VMX root mode and the TDX Module's SEAM Transfer VMCS (Virtual Machine Control Structure) is loaded.
- The SEAM Transfer VMCS is a means of switching the execution context between the hypervisor and the TDX Module.

Context Switches

■ TD ↔ TDX Module

- In traditional virtualization, the hypervisor handles VM exits, which are controlled by the VM's Transfer VMCS. Each VMCS is associated with one virtual CPU and stores the virtual CPU state for recovering the guest execution in the next VM resume.
- However, this operation leaks the virtual CPU state as VMCS is visible to hypervisors.
- In TDX, synchronous TDCALLs or asynchronous TD exits are designed to trap into the TDX Module. This is controlled by the TD Transfer VMCS, which is set up when a virtual CPU of a TD is created and stored in the TD's TDVPS.
- The TDVPS is encrypted with the TD's private key. Therefore, the **TD Transfer VMCS is inaccessible to untrusted hypervisors.**

Keyholes

- All memory buffers passed through SEAMCALLs use their physical addresses as references.
- The TDX Module must map these buffers into its own linear address space to access them.
- The Keyhole region is a reserved linear address range specifically for address mapping.